

LABORATORY OBSERVATION/RECORD

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 1

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Date: 29-08-2026

TASK 1 — HTML AND HTML5

1. TITLE

Develop a College Information Portal Using HTML and HTML5 Features

2. AIM

To understand and implement the fundamental building blocks of **HTML and HTML5** for creating a structured and interactive web page containing text, lists, tables, images, forms, semantic elements, multimedia, and HTML5 input types.

3. OBJECTIVE

The objectives of this task are:

- To understand the basic structure of an HTML5 document.
 - To create web pages using headings and paragraphs.
 - To apply different text-formatting tags.
 - To create ordered and unordered lists.
 - To add hyperlinks to external websites.
 - To create and display student details using tables.
 - To insert images with alternate text.
 - To design a student registration form.
 - To use HTML5 semantic elements.
 - To embed audio and video files.
 - To implement HTML5 input types such as email, date, number, color, and range.
 - To understand the role of HTML in structuring web content.
-

4. THEORY

HTML (HyperText Markup Language) is the standard markup language used to create and structure web pages. HTML uses elements and tags to define different parts of a web page such as headings, paragraphs, lists, tables, forms, images, and links.

HTML5 is the modern version of HTML that provides additional features such as semantic elements, multimedia support, canvas, and new form input types.

The basic structure of an HTML5 document consists of:

- `<!DOCTYPE html>` — Defines the document as an HTML5 document.
- `<html>` — Root element of the HTML document.
- `<head>` — Contains metadata and the page title.
- `<body>` — Contains the visible content of the web page.

Important concepts used in this task:

Headings and Paragraphs:

`<h1>` to `<h6>` are used for headings, while `<p>` is used for paragraphs.

Text Formatting:

`<b>` makes text bold, `<i>` makes text italic, `<u>` underlines text, `<sup>` displays superscript text, and `<sub>` displays subscript text.

Lists:

`<ul>` creates an unordered list, while `<ol>` creates an ordered list. Individual items are created using `<li>`.

Hyperlinks:

The `<a>` tag is used to create links to other web pages or websites.

Tables:

`<table>`, `<tr>`, `<th>`, and `<td>` are used to create and organize tabular information.

Images:

The `<img>` tag is used to display images. The `alt` attribute provides alternative text if the image cannot be displayed.

Forms:

HTML forms collect user information using controls such as text boxes, radio buttons, checkboxes, dropdowns, textareas, and buttons.

Semantic Elements:

HTML5 provides meaningful elements such as `<header>`, `<nav>`, `<section>`, `<article>`, and `<footer>` to describe the structure of a web page.

Multimedia:

The `<audio>` and `<video>` elements allow audio and video files to be embedded directly into web pages.

HTML5 Input Types:

HTML5 provides specialized input types such as email, date, number, color, and range, which provide better input handling and browser-level validation.

5. ALGORITHM / PROCEDURE

1. Create a new HTML file.
2. Define the document as an HTML5 document using `<!DOCTYPE html>`.
3. Create the basic `<html>`, `<head>`, and `<body>` structure.
4. Add the title **College Information Portal**.
5. Add headings and paragraphs containing college information.

6. Apply bold, italic, underline, superscript, and subscript formatting.
 7. Create an unordered list of available courses.
 8. Create an ordered list containing the application procedure.
 9. Add hyperlinks to the college website and Google.
 10. Create a student details table containing Roll No, Name, Branch, and Email.
 11. Insert the college logo and campus image using the tag.
 12. Add appropriate alternate text to the images.
 13. Create a student registration form.
 14. Add name, email, password, gender, hobbies, branch, comments, submit, and reset controls.
 15. Add HTML5 semantic elements such as header, navigation, section, article, and footer.
 16. Embed an audio file using the <audio> element.
 17. Embed a video file using the <video> element.
 18. Add HTML5 input types such as email, date, number, color, and range.
 19. Save the HTML file.
 20. Open the file in a web browser.
 21. Verify all elements, links, images, form controls, audio, and video.
 22. Record the output using screenshots.
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6. PROGRAM

index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>College Information Portal</title>
</head>
<body>
  <!--Headings and Paragraphs-->
  <h1>College Information Portal</h1>
  <h2>About Our Institute</h2>
  <p>Welcome to the official Information Portal</p>
  <p>Our college provides world-class education and fosters innovation among
students</p>
  <!--Text Formatting-->
  <h2>Academic Excellence</h2>
  <p>We are proud to be ranked<b> Number 1</b> in the region. Our<i> State-of-the-
art</i> campus offers a <u>dynamic learning environment</u>.</p>
  <p>Check out our award-winning formula :  $E=mc^2$ . Alternatively, look at our
chemistry lab schedules for  $H_2O$  solutions.</p>
  <!--Unordered List-->
  <h2>Available Courses</h2>
```

```

<ul>
  <li>Computer Science Engineerinh</li>
  <li>Data Science</li>
  <li>Artificial Intelligence</li>
</ul>
<!--Ordered List-->
<ol>
  <li>Fill out the online application form.</li>
  <li>Submit your Academic transcript.</li>
  <li>Attend the counseling session.</li>
</ol>
<!--HyperLinks-->
<h2>Useful Links</h2>
<p>Visit the official <a href="https://anits.org/" target="_blank">CollegeWebsite</a>for
more details.</p>
<p>For external research, you can search on <a href="https://www.google.com/"
target="_blank">Google</a>.</p>
<h2>Student Details</h2>
<!--Student details table-->
<table border="1">
  <tr>
    <th>Roll No</th>
    <th>Name</th>
    <th>Branch</th>
    <th>Email</th>
  </tr>
  <tr>
    <td>100</td>
    <td>Raj</td>
    <td>CSE</td>
    <td>raj@gmail.com</td>
  </tr>
  <tr>
    <td>101</td>
    <td>Priya</td>
    <td>CSM</td>
    <td>priya@gmail.com</td>
  </tr>
</table>
<!--Images-->
<h2>College Logo</h2>

<h2>Campus Image</h2>

<!--Forms-->
<h2>Student Registration Form</h2>
<form>

```

```

Name: <input type="text"><br><br>
Email: <input type="email"><br><br>
Password: <input type="password"><br><br>
Gender: <input type="radio" name="Gender">Male
<input type="radio" name="Gender">Female
<br><br>
Hobbies: <input type="checkbox">Sports
<input type="checkbox">Reading
<input type="checkbox">Music
<br><br>
Branch:
<select>
  <option>CSE</option>
  <option>ECE</option>
  <option>CSM</option>
  <option>CSD</option>
</select> <br><br>
Comments <br>
<textarea rows="4" cols="30"></textarea> <br><br>
<input type="submit">
<input type="reset">
</form>
<!--HTML5 Features-->
<header>
  <h1>College Information Portal</h1>
</header>
<nav>
  <a href="#">Home</a>
  <a href="#">About</a>
  <a href="#">Contact</a>
</nav>
<section>
  <article>
    <h2>About College</h2>
    <p>ANITS college provides quality education.</p>
  </article>
</section>
<h3>Audio</h3>
<audio controls>
  <source src="audio.mp3" type="audio/mpeg">
</audio>
<h3>Video</h3>
<video width="400" controls>
  <source src="video.mp4" type="video/mp4">
</video>
<form>
  Email: <input type="email"><br><br>

```

```

Date: <input type="date"><br><br>
Number: <input type="number"><br><br>
Favorite Color: <input type="color"><br><br>
Range: <input type="range"><br><br>
<footer>
    <p>Copyright 2026 ANITS College</p>
</footer>
</form>
</body>
</html>

```

7. CODE EXPLANATION

Code / Tag	Explanation
<!DOCTYPE html>	Declares the document as HTML5.
<html>	Defines the root element of the web page.
<head>	Contains metadata and page information.
<title>	Specifies the title displayed in the browser tab.
<h1>, <h2>, <h3>	Define different levels of headings.
<p>	Defines a paragraph.
	Displays text in bold.
<i>	Displays text in italic.
<u>	Underlines text.
<sup>	Displays superscript text.
<sub>	Displays subscript text.
	Creates an unordered list.
	Creates an ordered list.
	Defines an individual list item.
<a>	Creates a hyperlink.
<table>	Creates a table.
<tr>	Defines a table row.
<th>	Defines a table heading.
<td>	Defines table data.
	Displays an image.
alt	Provides alternative text for an image.
<form>	Creates an HTML form.
<input>	Creates different types of input controls.
<select>	Creates a dropdown list.
<textarea>	Creates a multi-line text input area.
<header>	Defines the header section of a page.
<nav>	Defines navigation links.
<section>	Defines a section of content.
<article>	Defines independent or self-contained content.
<footer>	Defines the footer section.

<audio>	Embeds an audio file.
<video>	Embeds a video file.
target="_blank"	Opens the linked page in a new browser tab.

8. TEST CASES AND EXECUTION DATA

The following test cases can be recorded after executing the webpage in a browser.

Test Case	Test / Input	Expected Result	Actual Result	Status
TC-01	Open index.html	College Information Portal should be displayed	College Information Portal displayed	Pass
TC-02	Check headings and paragraphs	Headings and college information should be displayed correctly	Displayed correctly	Pass
TC-03	Check text formatting	Bold, italic, underline, superscript and subscript should appear	Formatting displayed correctly	Pass
TC-04	Check ordered and unordered lists	Both lists should be displayed correctly	Lists displayed correctly	Pass
TC-05	Click College Website link	College website should open in a new tab	Link opened successfully	Pass
TC-06	Click Google link	Google should open in a new tab	Link opened successfully	Pass
TC-07	Check student table	Student details should be displayed in tabular form	Table displayed correctly	Pass
TC-08	Check logo and campus image	Images should be displayed with alternate text	Images displayed correctly	Pass
TC-09	Enter details in registration form	User should be able to enter the required information	Form controls worked correctly	Pass
TC-10	Check audio and video	Audio and video controls should be displayed	Multimedia displayed successfully	Pass
TC-11	Check HTML5 input types	Email, Date, Number, Color and Range controls should be available	Input types displayed correctly	Pass

9. OUTPUT

Output 1: College Information Portal

College Information Portal

About Our Institute

Welcome to the official Information Portal

Our college provides world-class education and fosters innovation among students

Academic Excellence

We are proud to be ranked **Number 1** in the region. Our *State-of-the-art* campus offers a dynamic learning environment.

Check out our award-winning formula : $E=mc^2$. Alternatively, look at our chemistry lab schedules for H_2O solutions.

Available Courses

- Computer Science Engineerinh
- Data Science
- Artificial Intelligence

1. Fill out the online application form.
2. Submit your Academic transcript
3. Attend the counseling session.

Useful Links

Visit the official [College Website](#) for more details.

For external research, you can search on [Google](#).

Student Details

Roll No	Name	Branch	Email
100	Raj	CSE	raj@gmail.com
101	Priya	CSM	priya@gmail.com

College Logo



Campus Image



Student Registration Form

Name:

Email:

Password:

Gender: ☐ Male ☐ Female

Hobbies: ☐ Sports ☐ Reading ☐ Music

Branch:

Comments

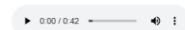
College Information Portal

[Home](#) [About](#) [Contact](#)

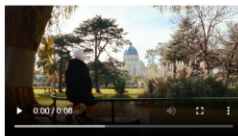
About College

ANITS college provides quality education.

Audio



Video



Email:

Date:

Number:

Favorite Color:

Range:

Copyright 2026 ANITS College

Figure 1: Output of the College Information Portal webpage showing headings, paragraphs, formatting, lists, hyperlinks, table, images, and form.

Supporting Files Used

The following files are used with the HTML page:

- logo.jpg — College logo
- campus.jpg — Campus image
- audio.mp3 — Audio file
- video.mp4 — Video file

logo.jpg



campus.jpg



10. INTERVIEW QUESTIONS

Q1. What is the purpose of semantic HTML5 elements?

Answer:

Semantic HTML5 elements provide meaningful information about the structure and purpose of different sections of a web page. Elements such as <header>, <nav>, <section>, <article>, and <footer> make the HTML code easier to understand, maintain, and interpret by browsers, search engines, and assistive technologies.

Q2. Compare Ordered and Unordered Lists.

Answer:

An **ordered list** () displays items in a specific sequence, usually using numbers or letters. An **unordered list** () displays items without indicating a particular order, usually using bullet points.

Q3. What happens if an image path is incorrect?

Answer:

If the image path is incorrect, the browser cannot find and display the image. Instead, the image will appear as broken or missing. The alternative text specified using the alt attribute may be displayed.

Q4. Differentiate Radio Buttons and Checkboxes.

Answer:

Radio buttons are generally used when the user must select **one option from a group of options**. Checkboxes are used when the user can select **multiple independent options**.

Q5. Explain the advantages of HTML5 input types over traditional text boxes.

Answer:

HTML5 input types provide specialized controls and built-in validation. For example, email validates email-format input, date provides a date selector, number supports numeric values, color provides a color picker, and range provides a slider. They improve usability and reduce the need for additional validation code.

11. RESULT

Thus, a **College Information Portal** was successfully developed using **HTML and HTML5**. Basic HTML elements, text formatting, lists, hyperlinks, tables, images, forms, semantic elements, multimedia, and HTML5 input types were implemented and verified successfully.

TASK 2 — HTML5 CANVAS AND CSS STYLING

1. TITLE

Implement HTML5 Canvas Graphics and Design a Styled Web Page Using CSS

2. AIM

To understand and implement **HTML5 Canvas** for drawing basic graphical elements and **CSS** for styling and designing a structured web page.

3. OBJECTIVE

The objectives of this task are:

- To understand the HTML5 <canvas> element.
 - To obtain the 2D drawing context using JavaScript.
 - To draw basic shapes using Canvas.
 - To draw a rectangle, circle, and line.
 - To display text using Canvas.
 - To understand external CSS styling.
 - To apply fonts, colors, spacing, borders, and shadows.
 - To create a navigation bar using CSS.
 - To design card-based content using Flexbox.
 - To apply hover effects to links and buttons.
 - To create a visually styled web page.
-

4. THEORY

HTML5 Canvas

The **HTML5 Canvas** element provides a drawing area that can be controlled using JavaScript. It can be used to create graphics, shapes, text, animations, and other visual elements.

The <canvas> element defines the drawing area, while JavaScript is used to perform the actual drawing.

The getContext("2d") method obtains the **2D drawing context**, which provides methods for drawing shapes and text.

In this task:

- fillRect() is used to draw a rectangle.
- arc() is used to create a circular path.
- stroke() is used to draw the line.
- fillText() is used to display text.
- fillStyle is used to set the fill color.
- font is used to define the text font.

CSS

CSS (Cascading Style Sheets) is used to control the appearance and layout of HTML elements.

External CSS can be connected to an HTML page using the `<link>` element.

In this task, CSS is used for:

- Font styling
 - Background gradient
 - Text color
 - Navigation styling
 - Flexbox layout
 - Cards
 - Borders
 - Rounded corners
 - Box shadows
 - Button styling
 - Hover effects
 - Footer styling
-

5. ALGORITHM / PROCEDURE

Part A — Canvas

1. Create an HTML5 document.
2. Add a `<canvas>` element with a width of 500 pixels and height of 300 pixels.
3. Assign an ID to the canvas.
4. Use JavaScript to access the canvas element.
5. Obtain the 2D drawing context using `getContext("2d")`.
6. Set the fill color for the rectangle.
7. Draw a rectangle using `fillRect()`.
8. Create a circular path using `arc()`.
9. Fill the circle using `fill()`.
10. Create a line using `moveTo()` and `lineTo()`.
11. Draw the line using `stroke()`.
12. Set the font and text color.
13. Display text using `fillText()`.
14. Open the HTML file in a browser and verify the output.

Part B — CSS Styled Web Page

1. Create an HTML5 document.
2. Add a heading inside the header.
3. Create a navigation bar containing Home, About, Services, and Contact links.
4. Create a container for the content cards.
5. Add separate cards for HTML, CSS, and JavaScript.
6. Add headings, descriptions, and buttons to each card.
7. Create a footer.
8. Create an external `style.css` file.
9. Link the CSS file to the HTML document.
10. Apply a gradient background to the page.
11. Style the header and navigation links.
12. Use Flexbox to arrange the cards.
13. Apply borders, rounded corners, and shadows to the cards.

14. Style the buttons.
 15. Add hover effects to navigation links and buttons.
 16. Open the HTML file in a browser.
 17. Verify the final styled webpage.
-

6. PROGRAM

Part A — HTML5 Canvas

Task2_PastA.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Task2</title>
</head>
<body>
  <canvas id="myCanvas" width="500" height="300" style="border:1px solid black;">
</canvas>

<script>

var c=document.getElementById("myCanvas");
var ctx=c.getContext("2d");

//Rectangle

ctx.fillStyle="navy";
ctx.fillRect(20,20,120,80);

//Circle

ctx.beginPath();
ctx.arc(220,60,40,0,2*Math.PI);
ctx.fillStyle="Pink";
ctx.fill();

//Line

ctx.moveTo(20,150);
ctx.lineTo(250,150);
ctx.stroke();
```

```
//Text
```

```
ctx.font="25px Arial";  
ctx.fillStyle="green";  
ctx.fillText("Your Name",20,220);
```

```
</script>  
</body>  
</html>
```

Part B — HTML Page

Task2_partB.html

```
<!DOCTYPE html>  
<html lang="en">  
<head>  
  <meta charset="UTF-8">  
  <meta name="viewport" content="width=device-width, initial-scale=1.0">  
  <title>My styled Webpg</title>  
  <link rel="stylesheet" href="style.css">  
</head>  
<body>  
  <header>  
    <h1>My Styled Web Page</h1>  
  </header>  
  <nav>  
    <a href="#">Home</a>  
    <a href="#">About</a>  
    <a href="#">Services</a>  
    <a href="#">Contact</a>  
  </nav>  
  <div class="container">  
    <div class="card">  
      <h2>HTML</h2>  
      <p>HTML Structures Web Pages</p>  
      <button>Read More</button>  
    </div>  
    <div class="card">  
      <h2>CSS</h2>  
      <p>CSS Styles Web Page</p>  
      <button>Read More</button>  
    </div>  
    <div class="card">  
      <h2>JavaScript</h2>  
      <p>JavaScript adds interactivity</p>
```

```
        <button>Read More</button>
    </div>
</div>
<footer>
    <p>Copyright 2026</p>
</footer>
</body>
</html>
```

style.css

```
body{
    font-family: Arial, Helvetica, sans-serif;
    margin: 0;
    background: linear-gradient(to right, #996b84,#336674);
}
header{
    color: white;
    text-align: center;
    padding: 10px;
    text-shadow: 2px 2px 5px black;
}
nav{
    padding: 10px;
    text-align: center;
}
nav a{
    color: white;
    padding: 10px;
    text-decoration: none;
    text-shadow: 2px 2px 5px black;
}
nav a:hover{
    background: #743348;
    border-radius: 5px;
}
.container{
    display: flex;
    justify-content: space-around;
    margin: 20px;
}
.card{
    background: white;
    padding: 20px;
    border: 2px solid gray;
    border-radius: 15px;
```

```

    box-shadow: 5px 5px 10px gray;
    width: 250px;
    text-align: center;
}
button{
    padding: 10px;
    border: none;
    background: rgb(16, 106, 124);
    color: white;
}
button:hover{
    background: rgb(222, 23, 76);
    color: white;
    text-align: center;
    padding: 15px;
}
footer{
    color: white;
    text-align: center;
    padding: 15px;
    text-shadow: 2px 2px 5px black;
}

```

7. CODE EXPLANATION

Part A — Canvas

Code	Explanation
<canvas>	Creates a drawing area on the web page.
id="myCanvas"	Provides an identifier for accessing the canvas using JavaScript.
width="500"	Sets the canvas width to 500 pixels.
height="300"	Sets the canvas height to 300 pixels.
getElementById()	Retrieves the canvas element from the HTML document.
getContext("2d")	Obtains the 2D drawing context.
fillStyle	Specifies the fill color of graphical elements.
fillRect()	Draws and fills a rectangle.
beginPath()	Starts a new drawing path.
arc()	Creates an arc or circular path.
fill()	Fills the current path.
moveTo()	Sets the starting point of a line.
lineTo()	Specifies the ending point of a line.
stroke()	Draws the current path.
font	Specifies the font and size of text.
fillText()	Displays text on the canvas.

Part B — HTML and CSS

Code	Explanation
<link rel="stylesheet">	Connects the external CSS file to the HTML page.
<header>	Defines the header section.
<nav>	Defines the navigation section.
.container	Defines the main container for the cards.
display: flex	Creates a flexible layout for arranging elements.
justify-content: space-around	Distributes the cards with space around them.
.card	Defines the appearance of individual content cards.
background	Sets the background color or gradient.
border	Adds a border around an element.
border-radius	Creates rounded corners.
box-shadow	Adds a shadow around an element.
text-shadow	Adds a shadow effect to text.
:hover	Applies styles when the mouse pointer moves over an element.
padding	Adds space inside an element.
margin	Adds space outside an element.
footer	Defines the footer section of the page.

8. TEST CASES AND EXECUTION DATA

Part A — Canvas Test Cases

Test Case	Test / Input	Expected Result	Actual Result	Status
TC-01	Open Canvas HTML page	Canvas area should be displayed	Canvas displayed correctly	Pass
TC-02	Execute rectangle code	A filled rectangle should appear	Rectangle displayed correctly	Pass
TC-03	Execute circle code	A filled circle should appear	Circle displayed correctly	Pass
TC-04	Execute line code	A line should appear on the canvas	Line displayed correctly	Pass
TC-05	Execute text code	Text should appear on the canvas	Text displayed correctly	Pass

Part B — CSS Test Cases

Test Case	Test / Input	Expected Result	Actual Result	Status
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TC-06	Open styled web page	Styled web page should be displayed	Displayed correctly	Pass
TC-07	Check header	Header should be centered and styled	Displayed correctly	Pass
TC-08	Check navigation links	Navigation links should be displayed horizontally	Displayed correctly	Pass
TC-09	Hover over navigation link	Hover background effect should appear	Hover effect displayed	Pass
TC-10	Check content cards	Three cards should be displayed using Flexbox	Cards displayed correctly	Pass
TC-11	Check card styling	Cards should have borders, rounded corners and shadows	Styling applied correctly	Pass
TC-12	Hover over button	Button hover effect should appear	Hover effect displayed	Pass
TC-13	Check footer	Footer should appear at the bottom with styled text	Footer displayed correctly	Pass

9. OUTPUT

Part A — HTML5 Canvas Output

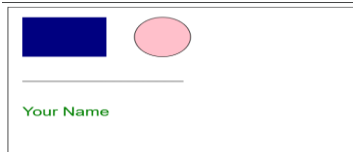


Figure 1: HTML5 Canvas output showing a rectangle, circle, line, and text.

Part B — CSS Styled Web Page Output

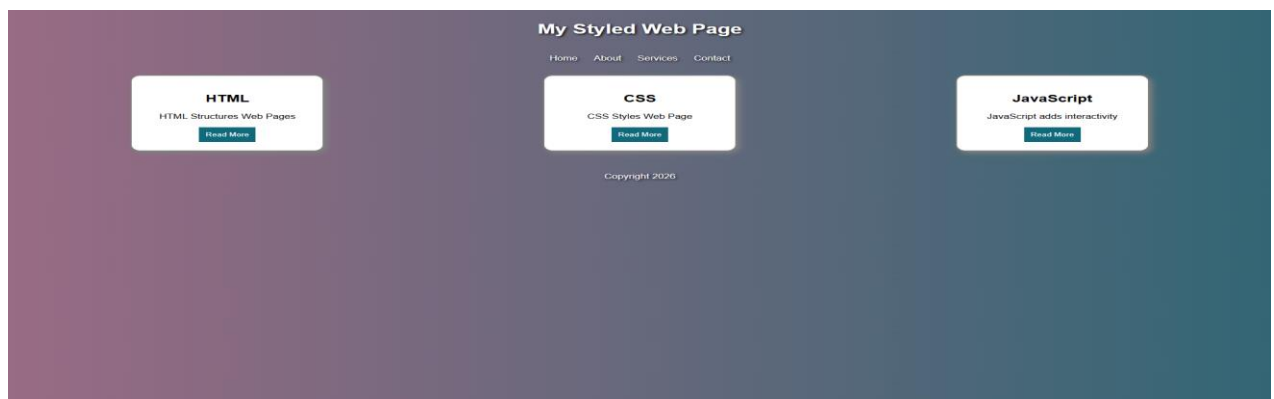


Figure 2: Styled web page showing header, navigation bar, HTML/CSS/JavaScript cards, buttons, and footer.

10. RESULT

Thus, **HTML5 Canvas graphics** were successfully implemented using JavaScript to draw a rectangle, circle, line, and text. A **styled web page** was also successfully developed using external CSS, including Flexbox layout, navigation styling, cards, borders, shadows, gradients, buttons, and hover effects. The outputs were verified successfully.